higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

ROADS
NQF LEVEL 2

22 NOVEMBER 2019

This marking guideline consists of 6 pages.

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QUESTION 1

1.1 1.1.1 **Cut operations**
A cut is to excavate or remove soil or earth,✓ to reach the required formation level✓ of the road.

1.1.2 **Fill operations**
A fill is when soil or earth is added✓ to reach the required formation level✓ of the road.

(2 × 2) (4)

1.2 1.2.1 'G' – Gravel and gravel/soil mixture.
1.2.2 'C' – Cementitious materials that used to strengthen gravel.
1.2.3 'A' – Asphalt surface material.
1.2.4 'S' – Bitumen surface treatment. (4 × 1) (4)

1.3
- Guardrail
- Rest stop
- Street light
- Road sign
- Traffic light
- Traffic cone
- Delineator
- Any other relevant road furniture (Any 2 × 1) (2)

[10]

QUESTION 2

2.1 2.1.1 **Gradient**
The vertical slope between two horizontal points✓
Vertical incline to a horizontal level.✓

2.1.2 **Centre line**
A point of reference established in the middle of a newly built road.✓
A guideline in the middle of a newly built road used to show the straight and curved alignment of the road.✓

(2 × 2) (4)

2.2
- String/string line (NOT gut line or fish line)
- Measuring tapes
- Pegs
- Hammer
- Builders lime
- Nails
- Shovel
- Rakes (Any 5 × 1) (5)

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2.3
- Through the slipping/sliding of the earth on which the slope occurs.
- By erosion of sand/soil layer.
- By increasing the moisture content of a clay soil.
- If the slope is not correct according to the engineering drawing. (4)

2.4
- Secure a string line at the given point (level) marked on the first profile.
- Stretch out and pull the string line across the road to the second profile.
- Hook and place a line level onto the string line in the middle between the two profiles.
- Move and adjust the string line at the second profile until the bubble in the vial is in the middle of the vial.
- Mark out the transferred point (level) on the second profile. (5)

2.5 2.5.1 **Benchmark**
Clearly marked point of reference on a construction site.

2.5.2 **"Mean level" of a benchmark**
Mean sea level

(2 × 1) (2)
[20]

QUESTION 3

3.1 3.1.1 **Gravel**
Particle size more than 2mm.
Consists of course particles taken from crushed rock.

3.1.2 **Sand**
Particles ranging from 0,06 mm to 2 mm.
Consists of granular particles taken from minerals and crushed rock.

3.1.3 **Clay**
Particle sizes of less than 0,002 mm.
Consists of fine grained particles formed from the erosion and weathering of rock.

3.1.4 **Silt**
Particle sizes of between 0,002 mm and 0,06 mm.
Consists of fine grains of soil minerals or humus material.

(4 × 2) (8)

3.2 3.2.1 **Permeability**
The ability of soil✓ to allow water to flow through soil.✓

3.2.2 **Compaction**
The process of reducing the soil volume by tightly pressing the soil particles closely together.✓
To improve the strength and stiffness of the road.✓

(2 × 2) (4)

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3.3
- To reduce dust during compaction.
- To make the soil particle stick together.
- To increase the strength of the soil.
- To produce and increase bearing capacity. (Any 3 × 1) (3)

3.4
- Dig up some of soil from the ground.
- Take a hand full of soil and squeeze it hand
- If the soil crumbles in your hand after squeezing it, the soil is too dry and if it leaves some moisture in your hand, the soil is too wet.
- After squeezing the soil, drop it to the ground.
- It the soil crumbles and breaks after dropping it to the ground, the soil is too dry. (5)

[20]

QUESTION 4

4.1
- Directs water away from the road surface.
- Improves driver safety on the road.
- Reduces erosion to the edge breaks of the road surface.
- Ensures the durability of the road.
- Increases the pavement lifetime of the road.
- Reduces maintenance and financial costs. (Any 5 × 1) (5)

4.2 Topsoil contains sufficient organic material, such as grass roots, seeds or pioneer plants,✓ which will help speed up the growth of the vegetation✓ on the slope surface or embankment. (2)

4.3 4.3.1 **Surface drainage**
- Collects water from the surrounding ground surfaces into side drains.
- Directs standing/stagnant water away from the road surface.
- Water flows away from a slope or designed gradient away from the road surface.
- Rapid drainage process/system collecting and diverting water in the event of floods or heavy rains.
- Channels water to a collection point, such as a stormwater drain.
- Usually constructed alongside the road. (Any 3 × 1) (3)

4.3.2 **Subsurface drainage**
- Collects rainwater that seeps through the soil from the ground surface into underground drainpipes.
- Collects water from sublevel tables into underground drainpipes.
- Water flows away from a slope or designed gradient away from the road pavement structure.
- Used in areas with excessive underground water.
- Usually constructed below ground level. (Any 3 × 1) (3)

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4.4 4.4.1 **Flexible culvert**

- Thin walled steel pipes
- Galvanized corrugated metal pipes
- Thick walled high density polyethylene (plastic) pipes

(Any 2 × 1)

4.4.2 **Rigid culvert**

- Reinforced concrete
- Cast iron
- Vitrified clay (Any 2 × 1)

(2 × 2) (4)

4.5

- Trench drains
- Herringbone drains
- Layer drains
- Structural drains (Any 3 × 1) (3)

[20]

QUESTION 5

5.1

- Earthworks below the formation level of the road.
- Earthworks above the formation level of the road. (2)

5.2

- Where excavation is dug in unstable soil.
- Heavy plant, equipment or materials too close to the sides of the excavation
- Where there is excessive vibration from vehicle traffic and construction equipment around the excavation.
- Water that has collected inside the excavation.
- Variations of different soil conditions.
- Other existing excavations next to or adjacent to the excavation.
- Where the soil is unable to support its own weight.
- Changes in the weather. (Any 6 × 1) (6)

5.3

- Must see to day-to-day operation of the facilities.
- Must cooperate and obey orders in the interest of safety.
- Must carry out regular inspections and maintenance of systems and equipment.
- Not purposely or otherwise endanger the health and safety of themselves or others.
- Not interfere with or abuse safety equipment. (Any 4 × 1) (4)

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5.4
- Excavator
- Back actor
- Bulldozer
- Dragline
- Front-end loader
- Tractor loader backhoe
- Grader
- Scraper
- Tipper truck (Any 5 × 1) (5)

5.5
- The road should be well drained with side drains in cases where there is excessive rainfall.
- The road should be as smooth as possible without large rocks.
- The road should not be too steep, to avoid slowing down the trucks.
- The road should be as straight as possible, without unnecessary bends.

(Any 3 × 1) (3)

5.6 The weight of the collapsing trench wall can twist the body causing injuries✓ or suffocate the person causing death.✓ (2)

5.7
- Excavating
- Hauling
- Spreading
- Compacting (Any 3 × 1) (3)

[25]

QUESTION 6

6.1
- Graded crushed stone or G1 to G3 materials.
- Crushed natural gravel or GS material.
- Water-bound macadam or WM to WM2 materials.
- Cement and natural stone/gravel or C1 to C4 materials. (Any ONE) (1)

6.2 6.2.1 **Earth road**
A road where no imported gravel is used,✓ but the roadway is formed, cleared and compacted by the flow of traffic.✓

6.2.2 **Gravel road**
A road with a layer of imported material✓ built to a specified width and standard.✓

(2 × 2) (4)

[5]

TOTAL: 100

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